

SQL Handbook (151–200 Commands with Before & After Tables - Exact Model)

151 GROUP CONCAT

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT GROUP_CONCAT(name) FROM students;

After:
Ravi,Kumar,Anu

152 COUNT DISTINCT WHERE

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT COUNT(DISTINCT city) FROM students WHERE age>19;

After:
2

153 SUM DISTINCT

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT SUM(DISTINCT age) FROM students;

After:
60

154 AVG DISTINCT

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT AVG(DISTINCT age) FROM students;

After:
20

155 CASE MULTI

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
SELECT CASE WHEN age>20 THEN 'High' WHEN age=20 THEN 'Mid' ELSE 'Low' END FROM students;
```

After:

Categorized

156 ORDER BY MULTI

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
SELECT * FROM students ORDER BY city, age DESC;
```

After:

Sorted output

157 LIMIT MULTI

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
SELECT * FROM students LIMIT 2 OFFSET 0;
```

After:

First 2 rows

158 EXISTS WHERE

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
SELECT * FROM students s WHERE EXISTS(SELECT 1 FROM students WHERE age>20);
```

After:

All rows

159 NOT IN

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:

```
SELECT * FROM students WHERE city NOT IN ('Chennai');
```

After:

Filtered

160 LIKE MULTI

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT * FROM students WHERE name LIKE '%a%';

After:
Names with 'a'

161 BINARY

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT BINARY 'Ravi';

After:
Binary string

162 CAST

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT CAST(age AS CHAR) FROM students;

After:
Converted

163 CONVERT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT CONVERT(age, CHAR) FROM students;

After:
Converted

164 JSON OBJECT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT JSON_OBJECT('name',name) FROM students;

After:

JSON output

165 JSON ARRAY

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT JSON_ARRAY(name,age) FROM students;

After:
JSON array

166 JSON EXTRACT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT JSON_EXTRACT('{\"a\":1}', '\$.a');

After:
1

167 WINDOW ROW_NUMBER

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT ROW_NUMBER() OVER() FROM students;

After:
1,2,3

168 WINDOW RANK

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT RANK() OVER(ORDER BY age DESC) FROM students;

After:
Ranked

169 DENSE RANK

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT DENSE_RANK() OVER(ORDER BY age DESC) FROM students;
After:
Dense rank

170 NTILE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore
SQL:
SELECT NTILE(2) OVER() FROM students;
After:
Buckets

171 LAG

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore
SQL:
SELECT LAG(age) OVER() FROM students;
After:
Previous row

172 LEAD

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore
SQL:
SELECT LEAD(age) OVER() FROM students;
After:
Next row

173 FIRST VALUE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore
SQL:
SELECT FIRST_VALUE(age) OVER() FROM students;
After:
First

174 LAST VALUE

Before Table (students):
| id | name | age | city |

1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT LAST_VALUE(age) OVER() FROM students;

After:
Last

175 CUME_DIST

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT CUME_DIST() OVER(ORDER BY age) FROM students;

After:
Distribution

176 PERCENT_RANK

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT PERCENT_RANK() OVER(ORDER BY age) FROM students;

After:
Percent

177 WINDOW SUM

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT SUM(age) OVER() FROM students;

After:
Total repeated

178 WINDOW AVG

Before Table (students):

id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT AVG(age) OVER() FROM students;

After:
Average

179 WINDOW COUNT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT COUNT(*) OVER() FROM students;

After:
Count
```

180 WINDOW PARTITION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
SELECT RANK() OVER(PARTITION BY city ORDER BY age) FROM students;

After:
Partition rank
```

181 CREATE USER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
CREATE USER 'u1'@'localhost' IDENTIFIED BY 'pass';

After:
User created
```

182 GRANT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
GRANT SELECT ON *.* TO 'u1'@'localhost';

After:
Granted
```

183 REVOKE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |

SQL:
REVOKE SELECT ON *.* FROM 'u1'@'localhost';

After:
```

Revoked

184 DROP USER

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
DROP USER 'u1'@'localhost';

After:
User removed

185 SHOW GRANTS

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW GRANTS FOR 'u1'@'localhost';

After:
Privileges

186 FLUSH

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
FLUSH PRIVILEGES;

After:
Refreshed

187 ALTER USER

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER USER 'u1'@'localhost' IDENTIFIED BY 'new';

After:
Updated

188 LOCK ACCOUNT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER USER 'u1'@'localhost' ACCOUNT LOCK;

After:
Locked

189 UNLOCK ACCOUNT

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER USER 'u1'@'localhost' ACCOUNT UNLOCK;

After:
Unlocked

190 SHOW PROCESSLIST

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SHOW PROCESSLIST;

After:
Processes

191 EXPLAIN ANALYZE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
EXPLAIN ANALYZE SELECT * FROM students;

After:
Plan

192 OPTIMIZE TABLE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
OPTIMIZE TABLE students;

After:
Optimized

193 ANALYZE TABLE

Before Table (students):
| id | name | age | city |

```
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
ANALYZE TABLE students;
```

After:

Stats updated

194 CHECK TABLE

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
CHECK TABLE students;
```

After:

Checked

195 REPAIR TABLE

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
REPAIR TABLE students;
```

After:

Repaired

196 SHOW STATUS

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW STATUS;
```

After:

Server stats

197 SHOW VARIABLES

Before Table (students):

```
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

SQL:

```
SHOW VARIABLES;
```

After:

Config

198 KILL QUERY

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
KILL QUERY 1;
```

```
After:  
Stopped
```

199 SHOW ENGINE

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
SHOW ENGINE INNODB STATUS;
```

```
After:  
Engine info
```

200 RESET QUERY CACHE

```
Before Table (students):  
| id | name | age | city |  
| 1 | Ravi | 20 | Chennai |  
| 2 | Kumar | 21 | Madurai |  
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:  
RESET QUERY CACHE;
```

```
After:  
Cache cleared
```